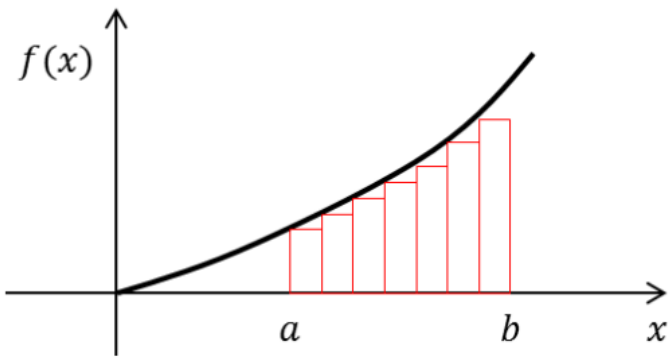
(16-C) Simulation Practice

(1번) Integrals using Monte Carlo Method

아래 그림은 적분의 개념을 보여주고 있다.

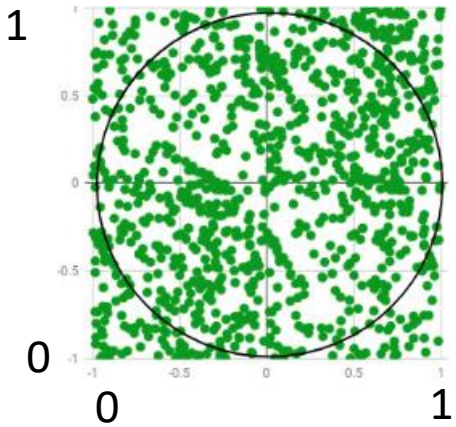


- 위의 그림을 고려하여 f(x) = x^5 - x^4 에 대해 닫힌 구간 [0, 10]의 적분값을 사각형의 갯수를 달리하여 Python으로 구해본다.
- 사각형 갯수는 10, 100, 1000, 10000, 100000, 1000000 을 적용하라.
- 실제 적분값은 다음과 같다.

$$\int_0^{10}(x^5-x^4)dx = \frac{440000}{3} \simeq 146667$$

(2번) Estimating Pi using Monte Carlo Method

- The idea is to simulate random (x, y) points in a 2-D plane with domain as a square of side 1 unit.
- Imagine a circle inside the same domain with same diameter and inscribed into the square.
- We then calculate the ratio of number points that lied inside the circle and total number of generated points.
- Refer to the image below:



We know that area of the square is 1 unit sq
while the size of circle is $\pi * (\frac{1}{2})^2 = \frac{\pi}{4}$

cir = number of points generated inside the circle
sq = number of points generated inside the square

$$\frac{cir}{sq} = \frac{\pi}{4} \quad \Rightarrow \quad \pi = 4 * \frac{cir}{sq}$$

- We simply generate random (x, y) pairs and then check if $x^2 + y^2 \leq 1$
- If yes, we increment the number of points that appears inside the circle.
- In randomized and simulation algorithms like Monte Carlo Method, the more the number of iterations, the more accurate the result is.
- Estimate Pi using the above discussion!